Candidate's Name:...……..……..

Index Number: ……Signature:…….

Total Weighted Score		
Initials		

545/1
CHEMISTRY
Paper 1
(THEORY)
July./ Aug.2026
2 hours

BOARD OF GOVERNORS EXAMINATION CONSULTS

Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES

*This examination paper has **two** Sections; **A** and **B**. it consists of **six** items*
*Section **A** has **two compulsory** items*

*Section **B** has **two** parts; **I** and **II**. Respond to **one** item from each part.*

*Respond to **four** examination items in all.*
Responses to section **B** *items* **must** *be written in the answer booklet(s) provided.*
*Any additional item(s) responded to will **not** be scored.*
Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

For Scorers's Use Only

Item No.	Basis Code	Weighted Score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
__(3/4)	P		
	S		
	N		
__(5/6)	N		
	M		
	Be		
Total Weighted Score			

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Turn Over

SECTION A

*Attempt **both** items in this section*

ITEM 1

The high demand for electricity in Uganda due to a large number of industries, growth increase, and the need for reliable power for home, hospital and factory use, scientists are exploring nuclear energy as an alternative source of electricity. During one of their discussions, they presented two nuclear reactions that release large amounts of energy;

Reaction **I** $^{239}_{94}Pu + {}^{1}_{0}n \rightarrow {}^{137}_{56}Xe + {}^{92}_{36}Kr + 3{}^{1}_{0}n + Energy$

Reaction **II** ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n + Energy$

However, it has been noted that only one of these nuclear reactions is commonly used to generate electricity at nuclear power stations while the other is capable of providing extremely large amounts of energy in the future.

Tasks

As a learner of chemistry;

a) Categorize the types of reactions **I** and **II**.

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b) Explain how one of the above reactions is used to generate electricity at nuclear power stations.

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c) Evaluate the use of the reactions above as a source of energy for generating electricity.

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d) Describe the environmental impacts that may arise from the use of nuclear energy.

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ITEM 2

During a practical chemistry lesson, the laboratory attendant guided the students to obtain the following data on three metals;

Element	**Melting Point (°C)**
Lithium	181
Sodium	98
Potassium	64

They further found out that exposure of these metals to air made them to form white solids, and combustion of one of the white solids formed by sodium showed that they contained 74.19% sodium.

Tasks

(a) Categorize the compounds formed when the metals are exposed to air.

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(b) State and explain the trend in the physical property of the metals in the table above.

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(c) Determine the empirical formula of the white solid formed when sodium is exposed to air.

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(d) Describe the environmental impacts of the compound formed in (c) above.

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SECTION B

PART I

*Respond to **only one** item from this part in the answer booklet(s) provided*

ITEM 3

To support the construction of dams, buildings and regional transport corridors in Western Uganda, an industrial consortium has proposed setting up a new Cement Processing Plant in Kasese District near the foots of Mt. Rwenzori.

While local leadership supports the plan, local tea farmers and residents are raising strong concerns about the environmental impact.



You have been chosen as a student representative on the District Environmental Assessment Committee meeting and asked to prepare a scientific presentation evaluating the cement factory's operations, impacts and address the residents' concerns.

Task

Write down detailed issues that you will present during the committee meeting.

ITEM 4

For many years, small-holder tea farmers in Kyenjojo District have experienced a severe drop in tea leaf yield and widespread yellowing of tea bushes (chlorosis) caused by sulfur and nitrogen deficiencies in the soil. To resolve this agricultural crisis and boost tea exports, a chemical firm has established an Ammonium Sulphate Fertilizer Manufacturing Plant in the district to supply local farmers with affordable nitrogenous and sulfur-rich fertilizer.

However, the establishment of the factory has sparked a heated debate among community members and residents while local youth leaders and traders support the

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factory development emphasizing job creation, improved household incomes and lower tea production costs.

You have been proposed by your school to present your opinions over the manufacture of fertilizers on a radio presentation.

Task

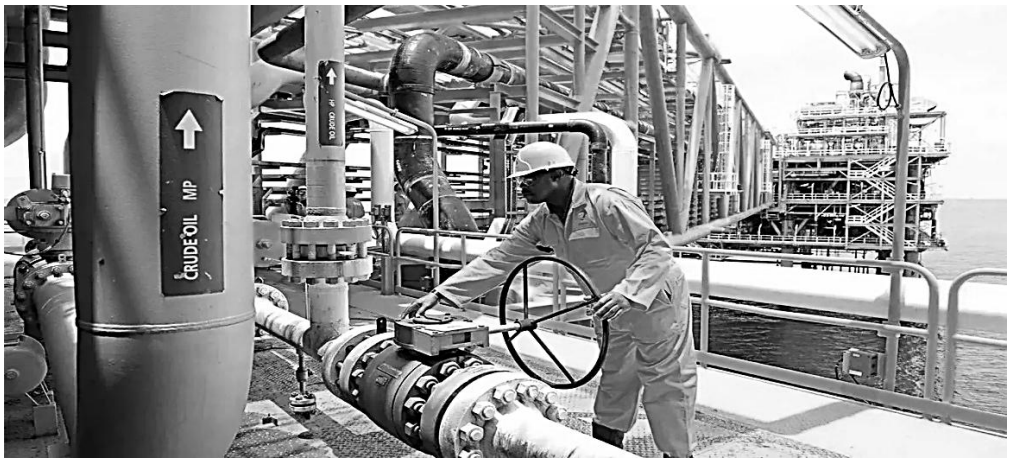
As a chemistry student write and explain key ideas you would discuss regarding the manufacture of ammonium sulphate in the district.

PART II

*Respond to **only one** item from this part in the answer booklet(s) provided*

ITEM 5

The discovery of crude oil in the Albertine region has led to increased oil mining, infrastructural development and better living standards in the area. Though, some residents have continuously complained that the activities carried out in the area are dangerous to the environment and local leaders have filed a lot of concerns from community members about the changes taking place around the oil mines.



Task

As a student who has studied natural resources and environmental conservation, your teacher asks you to prepare a presentation for your classmates on the issues surrounding crude oil exploitation in the Albertine region.

ITEM 6

The demand for coal is growing as a number of industries and homes depend on it as a good source of energy. However, it's extraction is linked to diverse challenges like air and water pollution as well as increased respiratory disorders which community members have noticed and complained or disagreed with those who benefit economically from coal extraction.

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Task

As a learner of chemistry, your class has been asked to discuss issues arising from the increased use of coal and exploitation of its resources. Write a detailed report of your presentation.

END

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